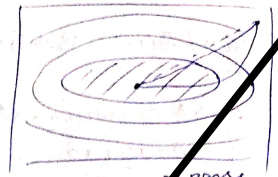


27/07/25 RMSprop - Root Mean Square Prop.

Improvement $\rightarrow$ AdaGrad $\leftarrow$ Sparse data.

AdaGrad: $V_t = V_{t-1} + (\nabla W_t)^2$

$$W_{t+1} = W_t + \frac{\eta}{\sqrt{V_t + \epsilon}} \nabla W_t$$

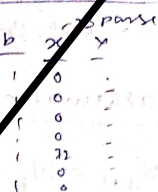


RMS Prop.

$$V_t = \beta V_{t-1} + (1-\beta) (\nabla W_t)^2$$

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t + \epsilon}} \nabla W_t$$

at the end $V_t \uparrow$
So the $\frac{\eta}{\sqrt{V_t + \epsilon}} \downarrow$ no update
So, cannot converge to the final minimum.



$\rightarrow V_0 = 0$

$\rightarrow V_1 = 0.95 \times 0 + 0.05 (\nabla W_1)^2$

$V_2 = 0.95 \times 0.05 (\nabla W_1)^2 + 0.05 (\nabla W_2)^2$

$V_3 = [0.95 \times 0.95 \times 0.05 (\nabla W_1)^2 + 0.45 \times 0.05 (\nabla W_2)^2] + [0.05 (\nabla W_3)^2]$

epoch 1

epoch 2

epoch 3

V_t short $\rightarrow \eta$

η not become very small.

$\rightarrow$ can converge.

largest.

AdaGrad $\rightarrow$ ~~LR~~ $\rightarrow$ ~~non~~ convex optimization.

(global minimum)

$\rightarrow$ prob. with non-convex $\rightarrow$ cannot converge.

but RMSprop $\rightarrow$ LR $\rightarrow$ NN $\} \times$

Disadvantage.

No, one of the best optimization techniques.

Adam Optimizer $\rightarrow$ Adaptive Moment Optimizer

SGD / BGD / MBGD

$\rightarrow$ Momentum $\rightarrow$ NAG
 $\rightarrow$ AdaGrad
 $\rightarrow$ RMS Prop.

$\rightarrow$ Momentum
 $\rightarrow$ learning decay $\rightarrow$ Merge

Mathematical Formulation

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t + \epsilon}} m_t$$

where $m_t = \beta_1 m_{t-1} + (1-\beta_1) \nabla W$ $\rightarrow$ Momentum

$V_t = \beta_2 V_{t-1} + (1-\beta_2) (\nabla W_t)^2$ $\rightarrow$ AdaGrad

Bias Correction

epoch #1 $\hat{m}_t = \frac{m_t}{1-\beta_1^t}$

$\hat{V}_t = \frac{V_t}{1-\beta_2^t}$

$\beta_1 = 0.9$
 $\beta_2 = 0.99$

change parameters in Keras.

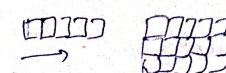
Verdict

Adam

$\rightarrow$ RMS Prop $\rightarrow$ Momentum.

28/07/25 CNN - Convolutional Neural Network

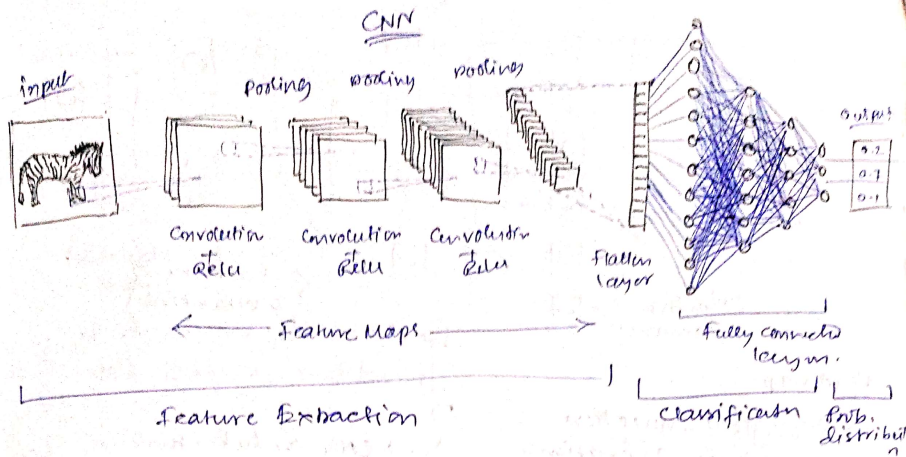
CNN - also known as convnet, are a special kind of NN for processing data that has a known grid-like topology like time series data (1D) or images (2D).



ANN $\rightarrow$ Convolution

ANN $\rightarrow$ Matrix Multiplication

$\rightarrow$ Convolution
 $\rightarrow$ Pooling
 $\rightarrow$ FC layer - ANN



Inspiration

→ Human Visual Cortex.

1998 → Yann Lecun → AT&T lab → to scan bank cheques.

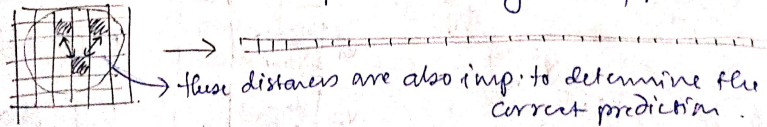
Later → Microsoft → OCR → handwriting Recognition } CNN.

{ facial recogn } CNN.
{ self driving } CNN. RNN.

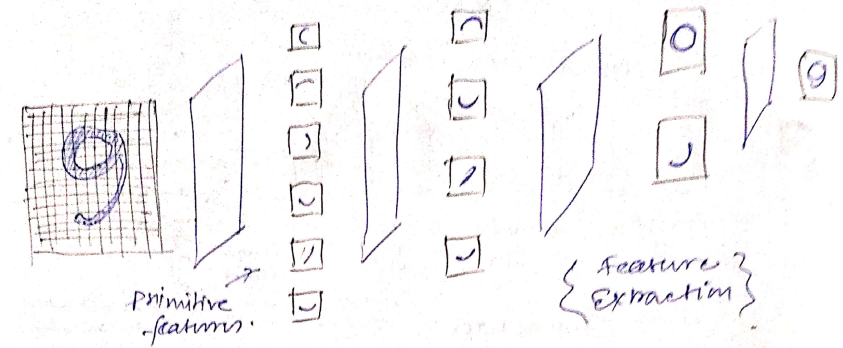
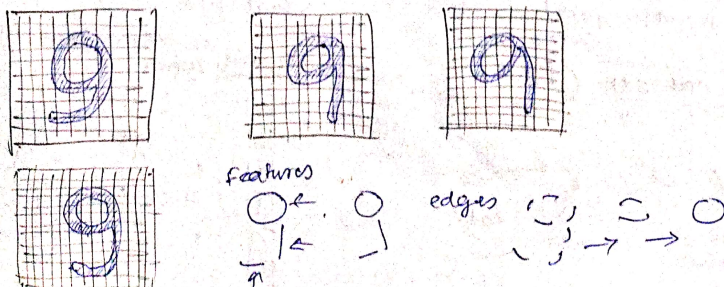
Why not use ANN? → on image.

MNIST → ANN → 98%.

1. High Computation Cost
2. Overfitting
3. Loss of imp info like spatial arrangement of pixels.

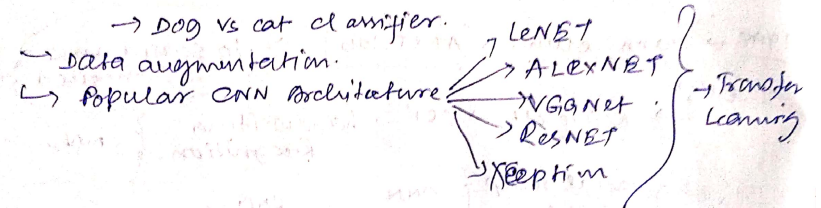


CNN intuition

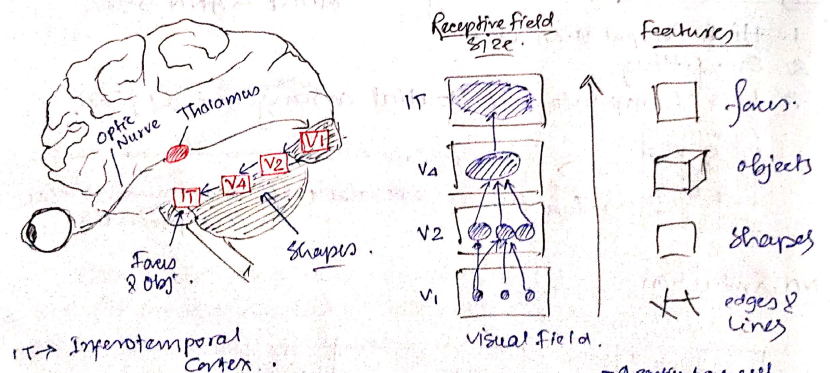


Roadmap

- Biological Connection
 - Convolution Operation.
 - ↳ padding + stride
 - Pooling layer
- CNN architecture.

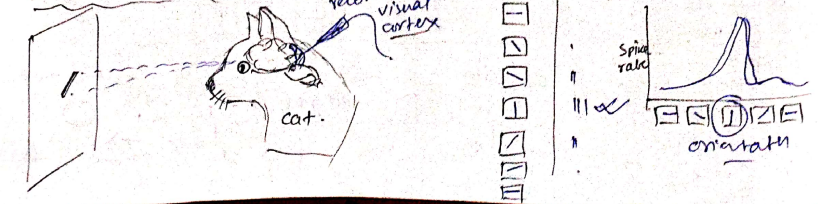


Human Visual Cortex

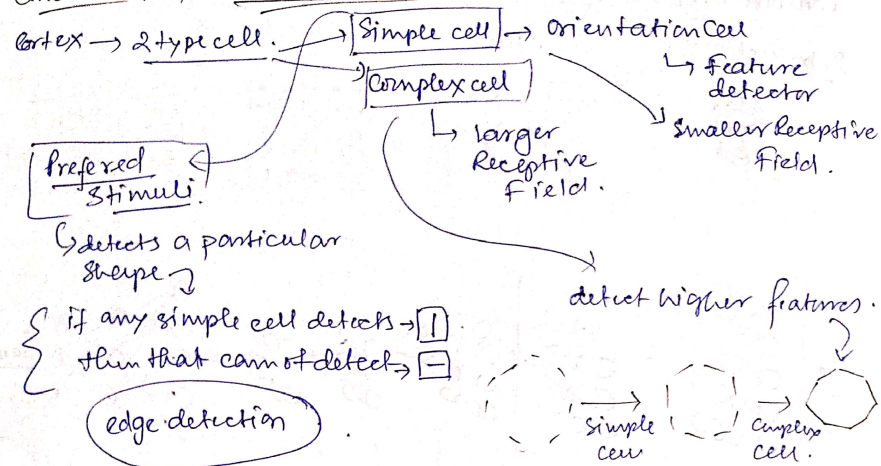


IT → Inferotemporal Cortex.

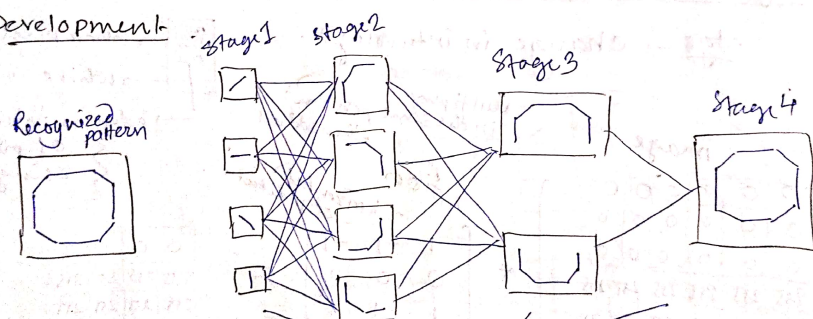
The cat exp. (1950-60)



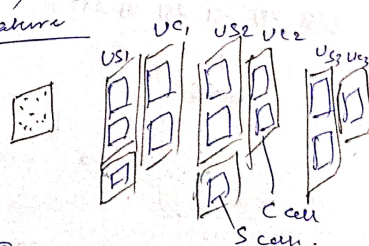
Cortex $\rightarrow$ 2 type cell.



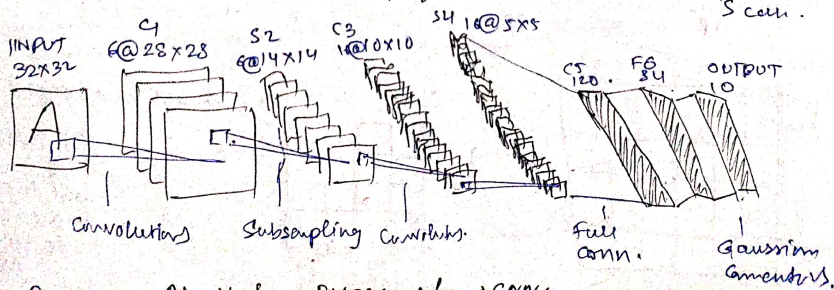
Development



Neocognitron $\rightarrow$ Japanese Scientist Fukushima.



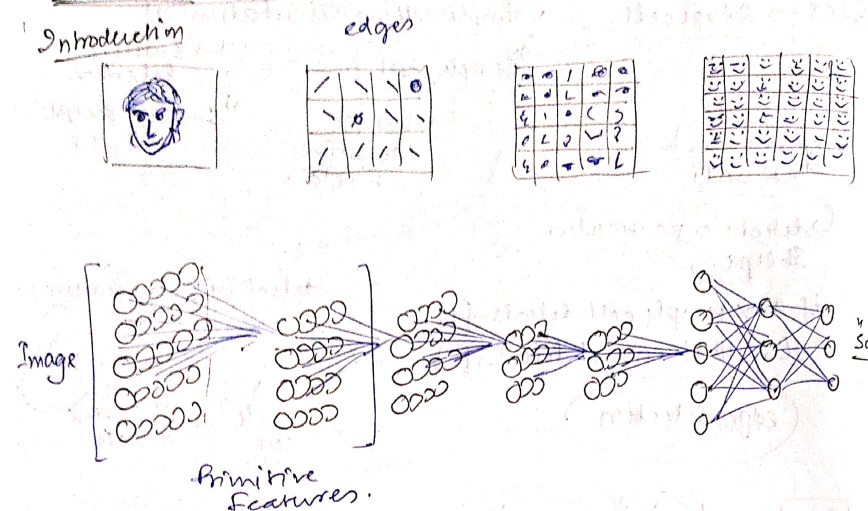
Yann Lecun $\rightarrow$ CNN $\rightarrow$ Backpropagation



2012 $\rightarrow$ AlexNet $\rightarrow$ ImageNet $\rightarrow$ CNNs

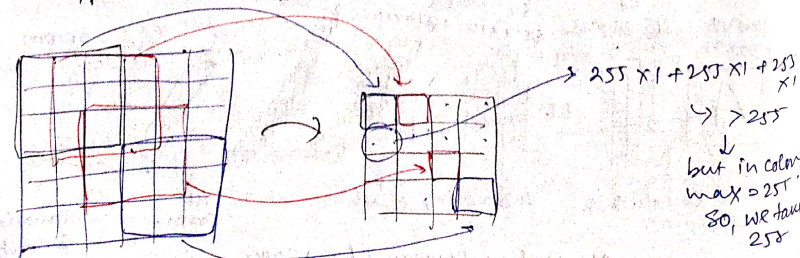
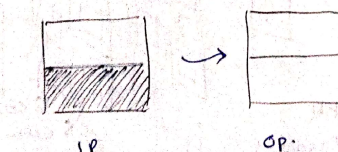
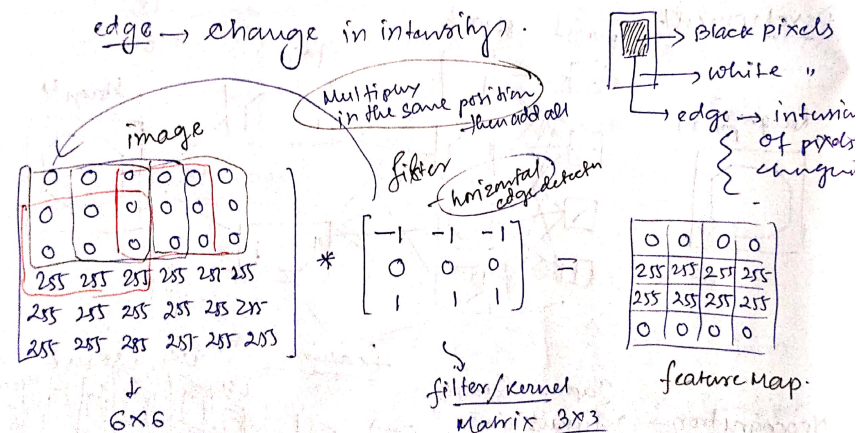
Convolution Operation

Introduction



Edge Detection (Convolution Operation)

edge $\rightarrow$ change in intensity.



Vertical edge detectn. → but in CNN, the values of the filter will automatically set while backpropagatn.

Grayscale.

$$\begin{matrix} \text{img} & * & \text{filter} & = & \text{result} \\ 6 \times 6 & & 3 \times 3 & & 4 \times 4 \\ (n \times n) & & (m \times m) & & \{(n-m+1) \times (n-m+1)\} \end{matrix}$$

RGB

$$\begin{matrix} \text{img} & * & \text{filter} & = & \text{result} \\ 6 \times 6 \times 3 & & 3 \times 3 \times 3 & & 4 \times 4 \end{matrix}$$

Filters are eyes of CNN.
value inside them decide which feature to detect

Multiple filters.

$$\begin{matrix} \text{img} & * & \text{Vertical Edge} & = & \text{result} \\ 6 \times 6 \times 3 & & 3 \times 3 \times 3 & & 4 \times 4 \\ & & \text{Horizontal Edge} & & \\ & * & 3 \times 3 \times 3 & = & 4 \times 4 \end{matrix}$$

4x4x2
filters used

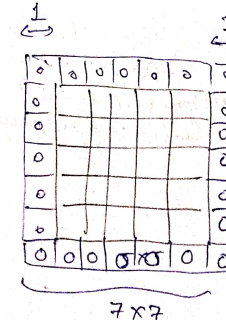
What is padding?

$$\begin{matrix} \text{img} & * & \text{filter} & = & \text{result} \\ n \times n & & f \times f & & (n-f+1) \times (n-f+1) \\ 5 \times 5 & & 3 \times 3 & & 3 \times 3 \end{matrix}$$

valid padding → no padding
same padding → i/p = o/p size

$n-f+1=3$
 $5-3+1=3$
 $n=5$

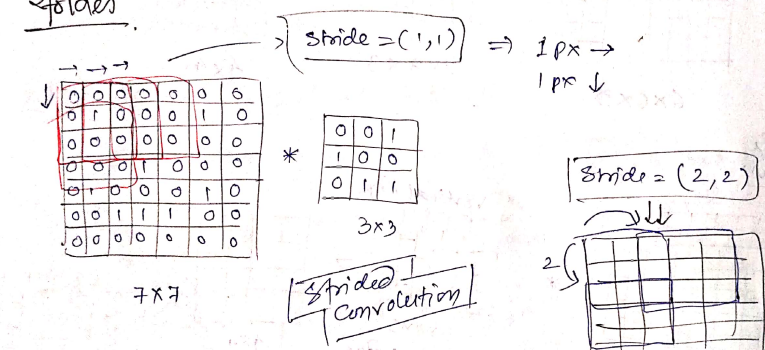
size reduced
→ the marginal pixels are not viewed multiple times.
if an image margin pixels contain imp. info. then this is not good for that.
so we have to increase the size.



zero padding.
padding performed to get the image with the same size as the original image.
 $(5 \times 5) \rightarrow (7 \times 7) * (3 \times 3) = (5 \times 5)$

with padding formula → $(n+2p-f+1)$
 $5+2 \times 1-3+1=5$

foldes



$$(n-f+1) \rightarrow \left\lceil \frac{n-f}{s} + 1 \right\rceil \rightarrow \text{padding} \rightarrow \left\lceil \frac{n+2p-f}{2} + 1 \right\rceil$$

stride

Special case s=2

$$\begin{matrix} \text{img} & * & \text{filter} & = & \text{result} \\ 8 \times 7 & & 3 \times 3 & & 2 \times 3 \end{matrix}$$

lack of pixels.

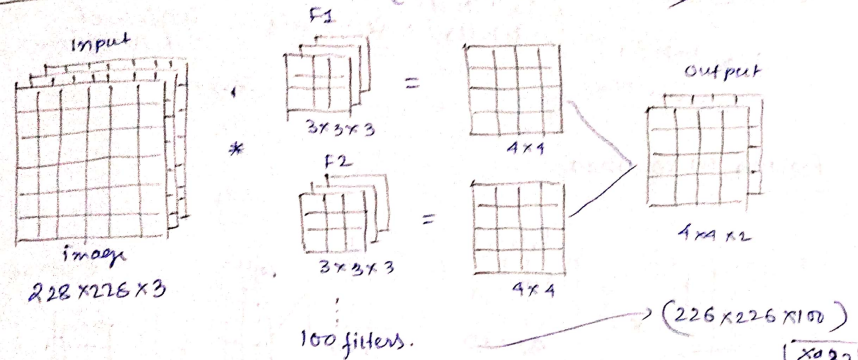
$n=8, f=3, s=2$
 $\frac{8-3}{2}+1=2.5+1=3$
 $\frac{7-3}{2}+1=2+1=3$
 $\therefore (3,3)$

$\frac{n-f}{s}+1=2$
 $\frac{6-3}{2}+1=1.5+1=2$
 $\therefore (2,3)$

Why Strides are required?

- 1) High level features
- 2) Computing.

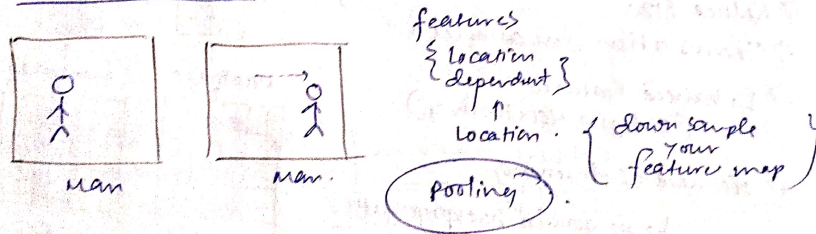
The problem with Strides. {
 1) Memory issue
 2) Translation variance



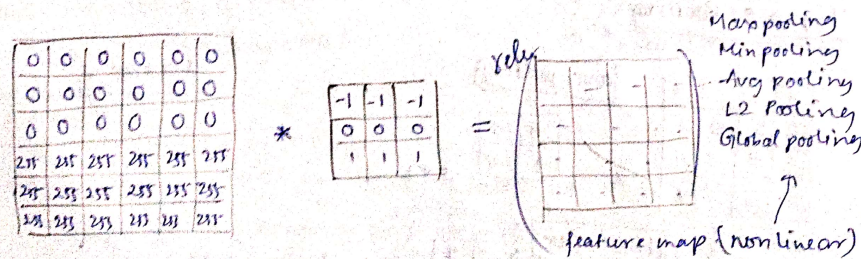
Memory issue -

Reduce size of feature map.
 Strides
 Pooling.
 Support batch size = 100
 19MB → for 1 training data.
 Total space = 19MB x 100 = 1900MB = 1.87GB ✓

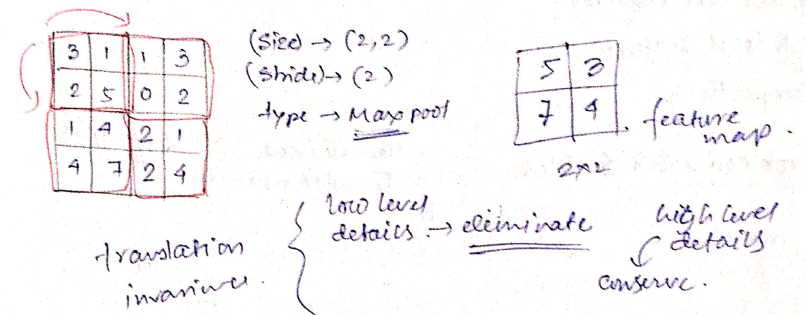
2) Translation Variance.



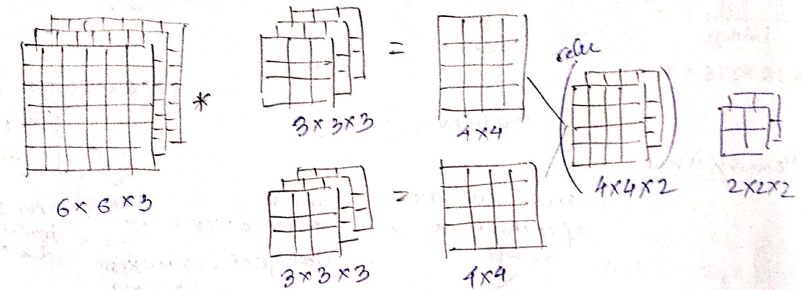
pooling



Pooling has no learnable parameters

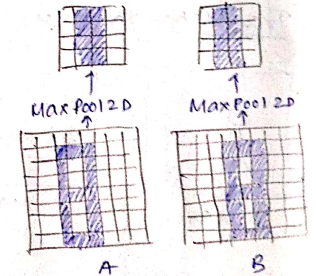


pooling on Volumes

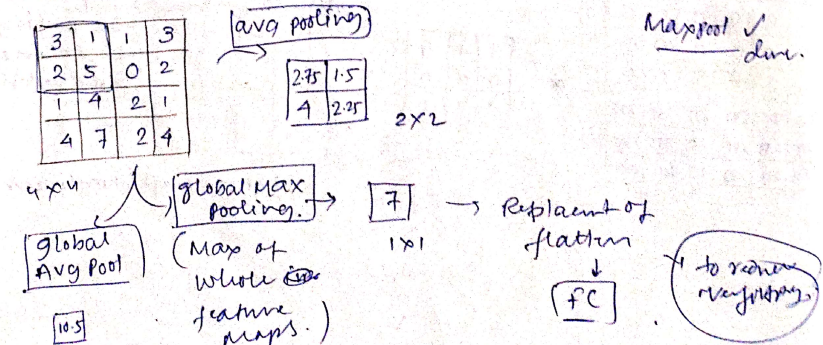


Advantages of pooling

- 1) Reduce Size
- 2) Translation Invariance
- 3) Enhanced features (only in Max Pooling)
- 4) No need of training
 ↳ No work in backpropagation.



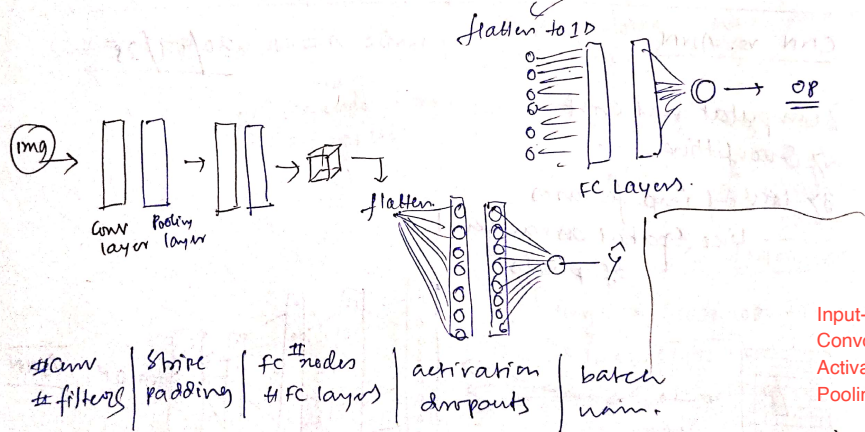
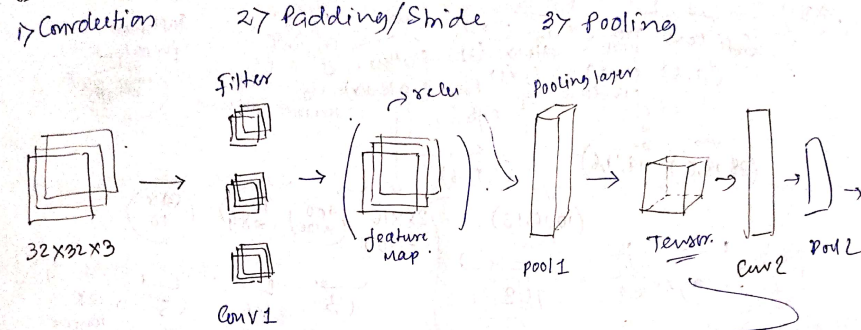
Types of Pooling



Disadvantages of Pooling

- NOT for Image Segmentation where every pixel matters
- loose information

CNN Architecture



#Conv | Stride | FC #nodes | activation | batch
#filters | padding | #FC layers | dropouts | norm.

ImageNET

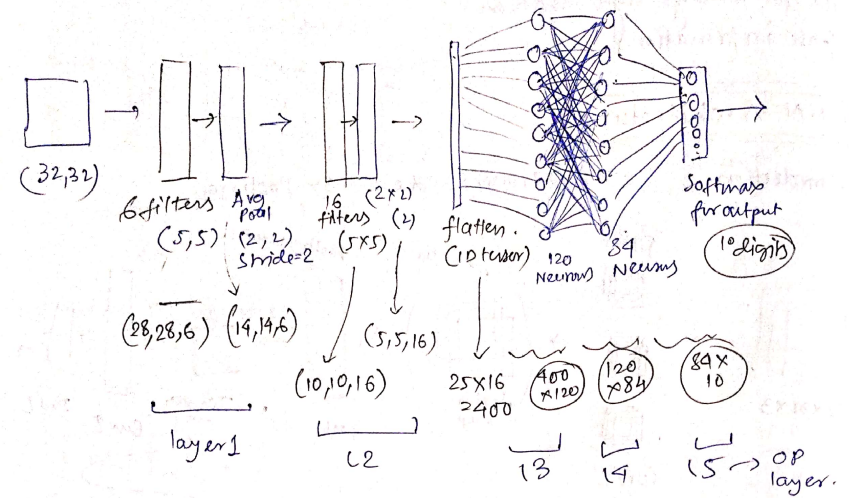
- 1) LeNET → Yann LeCun
- 2) AlexNet
- 3) GoogleNet
- 4) VGGNet
- 5) ResNet
- 6) Inception

LeNET

Yann Lecun → 1989 → 1998 → LeNET → CNN → US Navy Postal Structure

LeNET-5 → Because of 5 layers

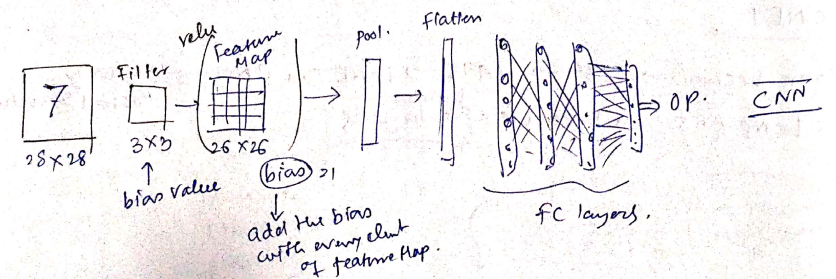
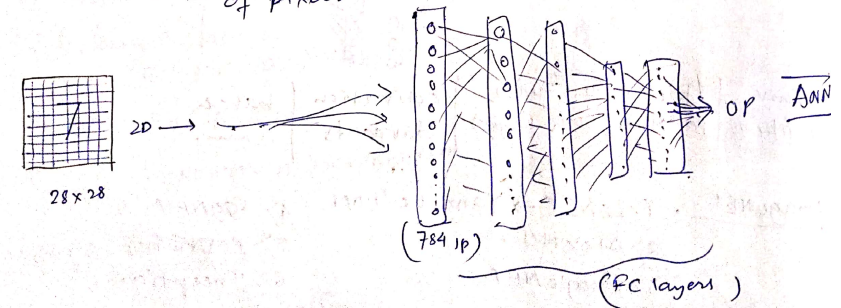
LeNET-5

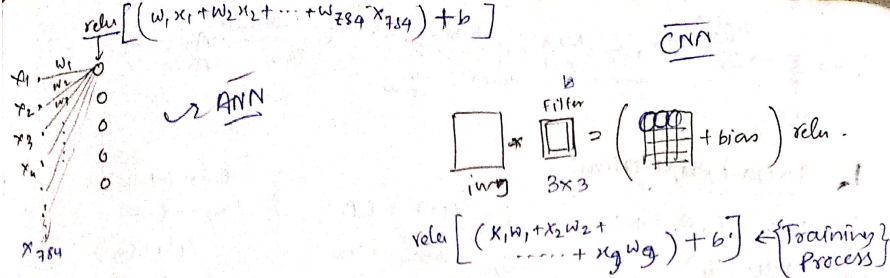


CNN vs. ANN

30/07/25

- 1) Computational cost
- 2) Overfitting
- 3) loss of imp features like spatial arrangement of pixels.



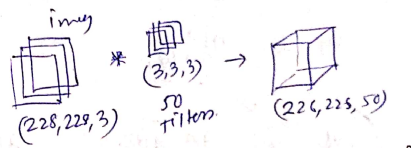


Operation likely similar $\Rightarrow$ ANN nodes $\approx$ CNN filters

Differences

1) if no. of input $\uparrow$
 $\hookrightarrow$ no. of wt $\uparrow$
 $\hookrightarrow$ ANN.
 but in CNN $\rightarrow$ no. of wt $\approx$ const
 $\downarrow$
 not dependent on no. of inputs.

Input size independence

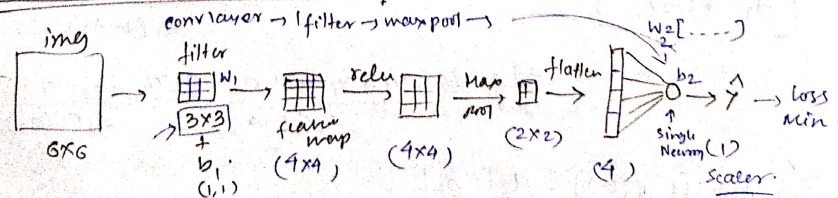


find out total learnable parameters b/w img & filters?

1 filter $(3, 3, 3) \rightarrow 3 \times 3 \times 3 = 27$ wts.
 $\times 50$
 $\rightarrow 1350$ wts.
 $+ 50$ bias
 $\rightarrow 1400$ w

if img size $(1080, 1080, 3)$??
 $\hookrightarrow$ Ans ≈ 1400 ✓
 (b/c η_w is not dependent on input)

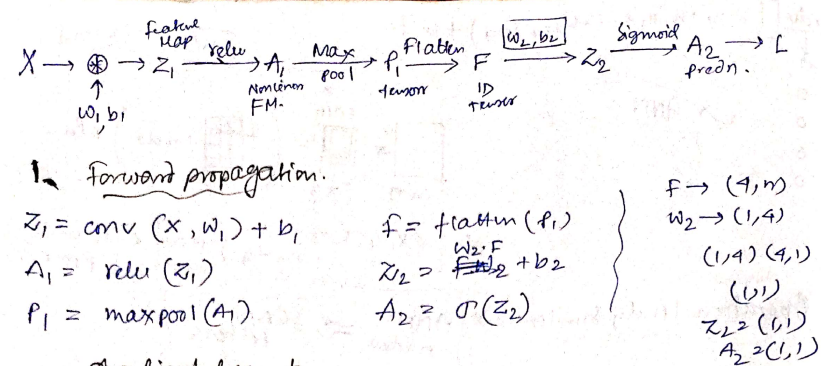
Backpropagation in CNN



Trainable parameters

$w_1 \rightarrow (3, 3)$ $w_2 \rightarrow (1, 1)$
 $b_1 \rightarrow (1, 1)$ $b_2 \rightarrow (1, 1)$

Loss $\rightarrow$ binary classification
 $L = -y_1 \log(\hat{y}_1) - (1-y_1) \log(1-\hat{y}_1)$
 $\Rightarrow$ total $9 + 4 + 1 + 1 = 15$ trainable parameters



gradient descent

$w_1 = w_1 - \eta \left[\frac{\partial L}{\partial w_1} \right]$
 $b_1 = b_1 - \eta \left[\frac{\partial L}{\partial b_1} \right]$
 $w_2 = w_2 - \eta \left[\frac{\partial L}{\partial w_2} \right]$
 $b_2 = b_2 - \eta \left[\frac{\partial L}{\partial b_2} \right]$

$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial F_1} \times \frac{\partial F_1}{\partial A_1} \times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial w_1}$
 $\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial F_1} \times \frac{\partial F_1}{\partial A_1} \times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial b_1}$
 $\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial F_1} \times \frac{\partial F_1}{\partial A_1} \times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial w_2}$
 $\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial A_2} \times \frac{\partial A_2}{\partial Z_2} \times \frac{\partial Z_2}{\partial F} \times \frac{\partial F}{\partial F_1} \times \frac{\partial F_1}{\partial A_1} \times \frac{\partial A_1}{\partial Z_1} \times \frac{\partial Z_1}{\partial b_2}$

Backprop

- convolution
- Flatten
- Maxpooling

F $\rightarrow$ ANN
 $\frac{\partial L}{\partial A_2} = \frac{\partial L}{\partial a_2} = \frac{\partial}{\partial a_2} [-y_1 \log(a_2) - (1-y_1) \log(1-a_2)]$
 $\frac{\partial L}{\partial A_2} = -\frac{y_1}{a_2} - \frac{(1-y_1)}{(1-a_2)} (-1)$
 $= -\frac{y_1}{a_2} + \frac{(1-y_1)}{(1-a_2)}$
 $= \frac{-y_1(1-a_2) + a_2(1-y_1)}{a_2(1-a_2)}$

$\frac{\partial L}{\partial A_2} = \frac{-y_1 + y_1 a_2 + a_2 - a_2 y_1}{a_2(1-a_2)} = \frac{a_2 - y_1}{a_2(1-a_2)}$

$\frac{\partial A_2}{\partial Z_2} = \sigma(Z_2) [1 - \sigma(Z_2)] = a_2 [1 - a_2]$

$\frac{\partial Z_2}{\partial F} = F$
 $\frac{\partial Z_2}{\partial b_2} = 1$

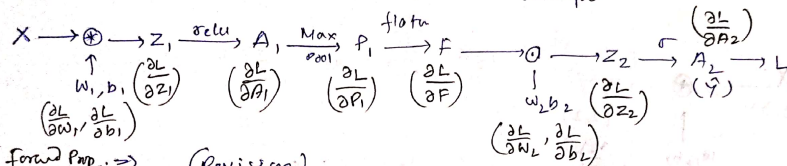
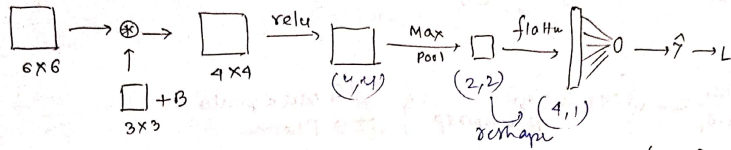
$\frac{\partial L}{\partial w_2} = \frac{(a_2 - y_1)}{a_2(1-a_2)} \times a_2(1-a_2) \times F = F(a_2 - y_1) = (A_2 - y) F^T$

$$\frac{\partial L}{\partial b_2} = \frac{(a_2 - y)}{a_2(1-a_2)} a_2(1-a_2) x_1 = (a_2 - y) - (a_2 - y)$$

$$\therefore \frac{\partial L}{\partial w_2} = (a_2 - y) F^T \quad \frac{\partial L}{\partial b_2} = (a_2 - y)$$

$$\begin{matrix} (1, m) - (1, m) \\ (1, m) \quad (m, 1) \\ \downarrow \\ (1, 1) \end{matrix}$$

21/07/25



Forward Pass $\Rightarrow$ (Revision)

$$Z_1 = \text{conv}(X, w_1) + b_1$$

$$A_1 = \text{relu}(Z_1)$$

$$P_1 = \text{max pool}(A_1)$$

$$F = \text{flatten}(P_1)$$

$$Z_2 = W_2 \cdot F + b_2$$

$$A_2 = \sigma(Z_2)$$

$$L = \frac{1}{m} \sum_{i=1}^m (-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i))$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial A_2} \cdot \frac{\partial A_2}{\partial Z_2} \cdot \frac{\partial Z_2}{\partial F} \cdot \frac{\partial F}{\partial P_1} \cdot \frac{\partial P_1}{\partial A_1} \cdot \frac{\partial A_1}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial w_1}$$

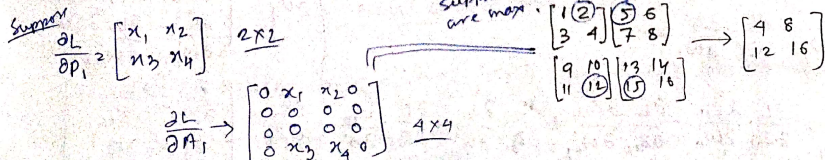
$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial A_2} \cdot \frac{\partial A_2}{\partial Z_2} \cdot \frac{\partial Z_2}{\partial F} \cdot \frac{\partial F}{\partial P_1} \cdot \frac{\partial P_1}{\partial A_1} \cdot \frac{\partial A_1}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial b_1}$$

Derivation

$$\frac{\partial Z_2}{\partial F} = W_2$$

$$\frac{\partial F}{\partial P_1} = \text{flatten}(P_1, \text{shape})$$

$$\frac{\partial L}{\partial w_1} = (A_2 - y) F^T \cdot w_2 \cdot \text{reshape}(P_1, \text{shape})$$



$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial A_2} \cdot \frac{\partial A_2}{\partial Z_2} \cdot \frac{\partial Z_2}{\partial F} \cdot \frac{\partial F}{\partial P_1} \cdot \frac{\partial P_1}{\partial A_1} \cdot \frac{\partial A_1}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial w_1}$$

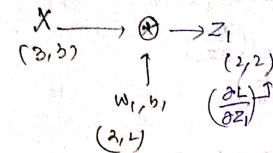
$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial A_2} \cdot \frac{\partial A_2}{\partial Z_2} \cdot \frac{\partial Z_2}{\partial F} \cdot \frac{\partial F}{\partial P_1} \cdot \frac{\partial P_1}{\partial A_1} \cdot \frac{\partial A_1}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial b_1}$$

$$\frac{\partial L}{\partial A_1} = \begin{cases} \frac{\partial L}{\partial P_1} & \text{if } A_{1,xy} \text{ is the maximum element} \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{\partial A_1}{\partial Z_1} = \begin{cases} 1 & \text{if } Z_{1,xy} > 0 \\ 0 & \text{if } Z_{1,xy} < 0 \end{cases}$$

$$\frac{\partial Z_1}{\partial w_1} = \left\{ \begin{array}{l} \text{convolution} \\ \text{Backprop} \end{array} \right\} \rightarrow \begin{array}{l} \text{max pooling} \\ \text{flatten} \end{array}$$

In convolution,



$$Z_1 = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

$$W_1 = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

$$Z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{21}w_{21} + x_{22}w_{22}$$

$$Z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{22}w_{21} + x_{23}w_{22}$$

$$Z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{31}w_{21} + x_{32}w_{22}$$

$$Z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{32}w_{21} + x_{33}w_{22}$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial b_1} = \frac{\partial L}{\partial Z_{11}} \cdot \frac{\partial Z_{11}}{\partial b_1} + \frac{\partial L}{\partial Z_{12}} \cdot \frac{\partial Z_{12}}{\partial b_1} + \frac{\partial L}{\partial Z_{21}} \cdot \frac{\partial Z_{21}}{\partial b_1} + \frac{\partial L}{\partial Z_{22}} \cdot \frac{\partial Z_{22}}{\partial b_1}$$

$$= \frac{\partial L}{\partial Z_{11}} + \frac{\partial L}{\partial Z_{12}} + \frac{\partial L}{\partial Z_{21}} + \frac{\partial L}{\partial Z_{22}} = \text{sum}\left(\frac{\partial L}{\partial Z_1}\right)$$

$$\frac{\partial L}{\partial b_1} = \text{sum}\left(\frac{\partial L}{\partial Z_1}\right) \rightarrow \text{scalar}$$

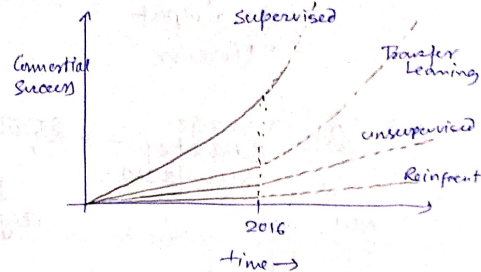
$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial Z_1} \cdot \frac{\partial Z_1}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial Z_{11}} \cdot \frac{\partial Z_{11}}{\partial w_{11}} + \frac{\partial L}{\partial Z_{12}} \cdot \frac{\partial Z_{12}}{\partial w_{11}} + \frac{\partial L}{\partial Z_{21}} \cdot \frac{\partial Z_{21}}{\partial w_{11}} + \frac{\partial L}{\partial Z_{22}} \cdot \frac{\partial Z_{22}}{\partial w_{11}}$$

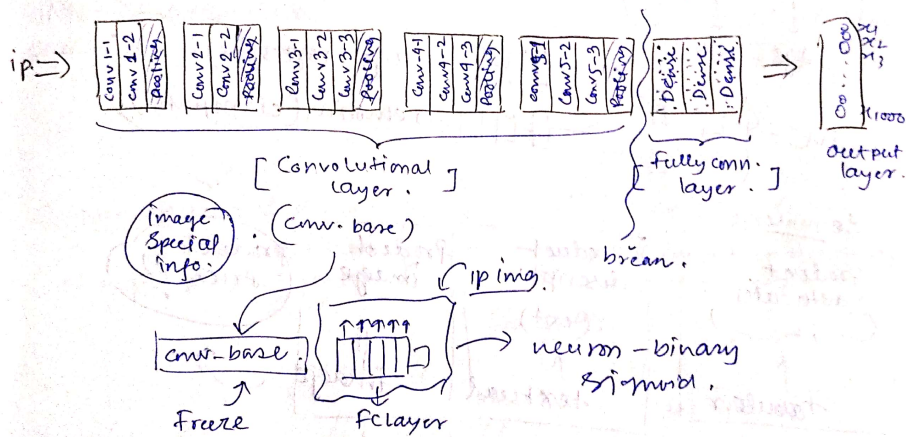
$$\frac{\partial L}{\partial w_1} = \begin{bmatrix} \frac{\partial L}{\partial w_{11}} & \frac{\partial L}{\partial w_{12}} \\ \frac{\partial L}{\partial w_{21}} & \frac{\partial L}{\partial w_{22}} \end{bmatrix}$$

Transfer Learning

Transfer Learning is a research problem in ML that focuses on storing knowledge gained while solving one problem and applying on it for a different but related problem.



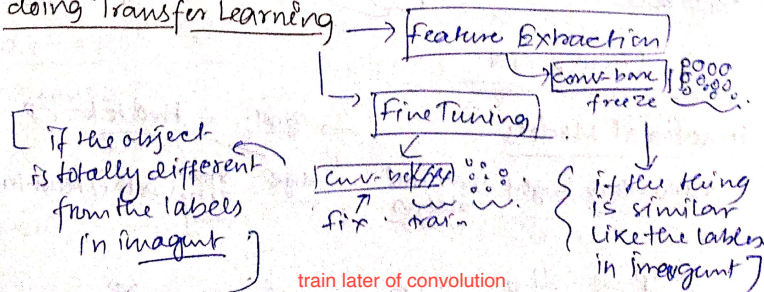
VGG 16 Model Architecture



Why transfer Learning works

The primitive features are already learned by the model in the conv. base. So, no need to train them again.

Days of doing Transfer Learning



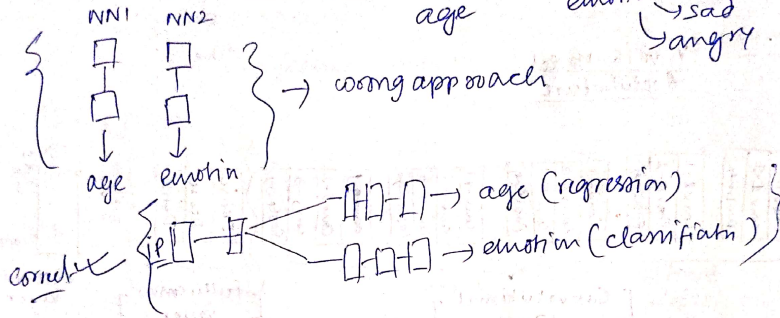
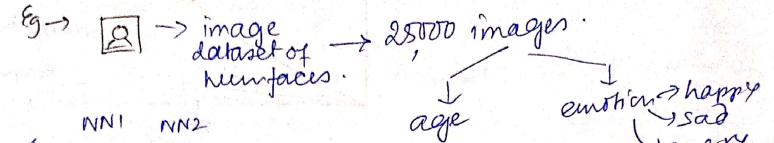
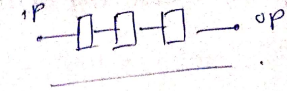
feature extraction then fine tuning

Problem With Sequential Model

ANN → CNN

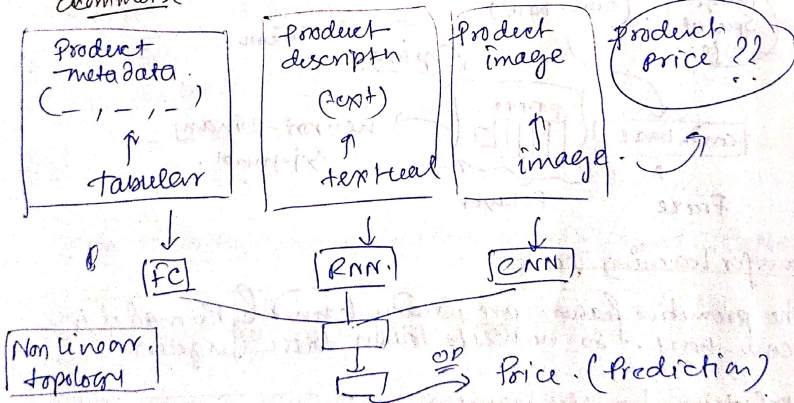
→ Sequential - keras
→ 1 input
→ 1 output
→ Linear

Linear topology



Branching Structure

Ecommerce



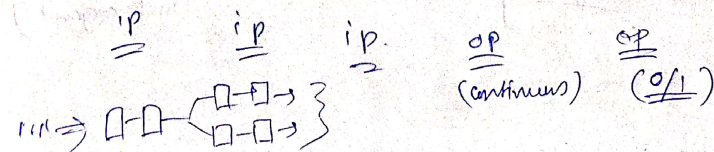
Functional API

Non Linear NN

Functional Models

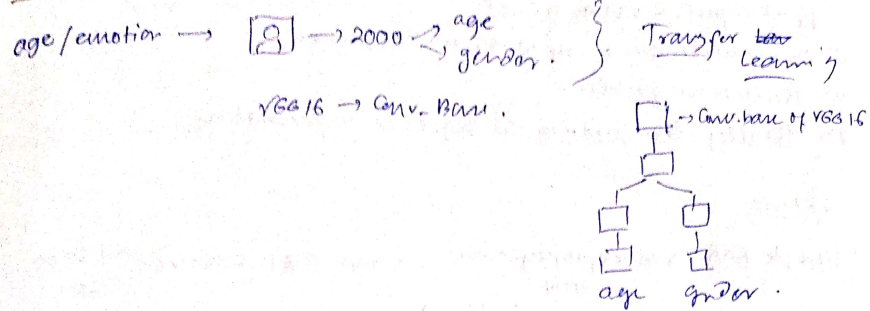
yearly salary height marital status

→ 301. Predict??
age stat city (Delhi/Mumbai)



only custom FC

Multi Output Model



RNN — Recurrent Neural Network

ANN → tabular data
CNN → images/grid.

RNN → is a type of sequential model to work on sequential data.

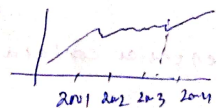
iq marks gender placent ?

Non sequential data.
if iq & marks position changed placent do not affect.

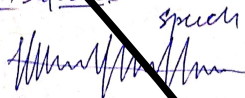
text data

Hi My name is Anik ✓
Hi Anik is name my. X

Time Series



DATA sequence



Why use RNN?

ip	op
Hi my name is Anik (5)	0
I love complex (3)	0
India won the match (4)	1

Vectorize (one-hot)

Hi My name.
[1, 0, 0, ..., 0] [0, 1, 0, ..., 0] ... [0, 0, 1, 0, ..., 0]

Anik.
[0, 0, 0, 0, 1, ..., 0]

ANN → text classification

ip1	ip2
12	0
12	0 0
12	0 0
12	0 0
12	0 0

this input layers vary
different ip size.

Problems.

- 1) Text input → varying size
- 2) Zero padding → unnecessary computation
- 3) Prediction problem.
- 4) Totally disregarding the sequence information.

Roadmap

Simple RNN → Backpropagation → RNN. → Solve LSTM GRU.
(Backprop in time)

Types of RNN

Seq. Model

Bidirectional RNN

Deep RNN

4/08/25

Why RNN?

1) Sequence → any length → ANN → fixed ip size
movie review

2) Sequence contains some meaning.

ANN
↑
memory feature

Data for RNN

(timesteps, input-features)

Review	Sentiment
Movie was good	1
movie was bad	0
movie was not good	0

review1
[[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0]]

(3, 5)
No. of timesteps
No. of features

review2 → (3, 5)
review3 → (4, 5)

Vocab → 5 words
Movie was good
[1, 0, 0, 0] [0, 1, 0, 0] [0, 0, 1, 0]
bad not good

Keras → Simple CNN →
(batch size, timestep, input)